

CLOUD COMPUTING LAB 01

Exercise 1

Amazon Elastic Compute Cloud (EC2) instances typically use hardware virtualization, specifically Xen hypervisor-based virtualization. However, Amazon also offers alternative virtualization options, such as:

Hardware Virtual Machine (HVM): This is the most common type of virtualization used by EC2 instances. HVM virtualization provides full hardware virtualization, allowing EC2 instances to run a wide range of operating systems without modification. This type of virtualization allows for better isolation between instances and provides the flexibility to run various operating systems, including Windows, Linux, and others.

Paravirtual (PV): In this type of virtualization, the guest operating system is modified to run on the Xen hypervisor without requiring full virtualization of the hardware. However, PV instances have become less common compared to HVM instances.

Nitro Hypervisor: Amazon introduced the Nitro Hypervisor for newer generation instances. It offloads many virtualization functions to dedicated hardware components, providing better performance and security isolation. This hypervisor is used in conjunction with HVM instances.

The specific type of virtualization used by an EC2 instance depends on factors such as the instance type, the underlying hardware, and the chosen AMI.

In this lab we used Hardware Virtual Machine (HVM) as you can see below

Instance: i-0bb7fbc3154f0ca53 (ubuntu)

► Instance summary [Info](#)

► Instance details [Info](#)

▼ Host and placement group [Info](#)

Host ID	–	Affinity	–
Host resource group name	–	Tenancy	 default
Virtualization type	 hvm	Reservation	 r-0bd067936c310d48a
Number of vCPUs	1		